

Jihoon Chung, Ph.D., LEED AP BD+C, WELL AP, FMP

CONTACT INFORMATION

Grid-interactive Controls Group
Oak Ridge National Laboratory
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QUALIFICATION SUMMARY

- Interdisciplinary researcher specializing in Building Information Modeling (BIM), Internet of Things (IoT), and ontology, and occupant behavior modeling for enhancing indoor environmental quality and energy efficiency
- Held a Ph.D. in Architectural Science at Rensselaer Polytechnic Institute, focusing on developing digital twin platforms for co-simulation ecosystem for multidisciplinary simulation environments
- Developed web-based digital twin platforms integrating BIM, IoT, and Building Automation Systems (BAS) for real-time monitoring of indoor environment and occupant behaviors, leading to publications in i3CE and ICCCBE proceedings
- Collaborated with industry and academic leaders, including Autodesk, Lawrence Berkeley Lab, NASA, Solibri, Tetherless World Constellation (TWC), LESA laboratory, and OBMI
- Experienced educator and mentor, serving as a guest lecturer, project mentor, and teaching assistant in programming and smart IoT systems at Rensselaer Polytechnic Institute and high schools in NYC and San Francisco

EDUCATION

Rensselaer Polytechnic Institute, New York, United States
Ph.D., Architectural Sciences (Built Ecologies), GPA: 4.0/4.0, May 2025

- Thesis: “Cross Domain Interoperability Framework for Enhancing Building Performance: Integrating Building Information Modeling, Building Energy Modeling, Building Control System, and Occupant Behavior”
- Advisor: Dr. Dennis Shelden

Graduate Certificates, May 2024

- Machine Learning & AI | Business Intelligence

M.S., Architecture (Design Technology), GPA: 4.0/4.0, December 2023

- Advisor: Dr. Dennis Shelden

Yonsei University, Seoul, Korea
M.S., Architectural Engineering (Construction IT), GPA: 3.97/4.0, February 2021

- Thesis: “Technical Specification Framework for 3D Concrete Printers”
- Advisor: Dr. Ghang Lee

B.Arch, Architecture (5–year program), February 2019

National University of Singapore, Singapore
Exchange Program for a year in AY2017

Gunmyungwon (건명원), Seoul, Korea
Future New School of Humanities, Arts & Science, February 2016

RESEARCH & WORK EXPERIENCE

Oak Ridge National Laboratory, Tennessee
Postdoctoral Research Associate **May 2025 – Current**

- Developed scalable building control infrastructure across buildings and campuses for grid-interactive control and optimization, integrating advanced control strategies such as Model Predictive Controls

- Grid-interactive Controls Group (led by Dr. Jamie Lian) under Energy Science and Technology Directorate
- Served as a Budget & Sponsorship Chair at ORPA Executive Committee Board for FY26

Oak Ridge National Laboratory, Tennessee

February 2025 – May 2025

ORISE Graduate Researcher

- Developed an open-source, generalizable building control web application for small- and medium-sized commercial buildings by integrating VOLTTRON backend and building control emulator BOPTEST
- Participated in *Graduate Research at ORNL* program (Mentored by Dr. Sen Huang)

Center for Architecture Science and Ecology, New York

August 2021 – May 2025

Research Assistant

- Developed prototypes of a SHACL-based IFC-to-Brick converter and an IFC4.3-to-ECS translator in collaboration with industry experts from Autodesk, Solibri, and Trimble, showcased them at the IFC Implementers Meeting of buildingSMART International
- Developed a web-based building monitoring system for occupant behavior using BIM, IoT, Building Automation System (BAS), and semantic web technologies
- Designed Arduino circuit boards and developed a web server for IoT-based plant modules to autonomously manage plant health by controlling water pumps, grow lights, and fans, as a part of Plant Pixel project sponsored by OBMI
- Organized and hosted the guest lecture series “AI Technology in Built Ecologies” as part of the ‘CASE Built Ecologies Graduate Seminar’, featuring four expert lecturers specializing in AI technologies within the field of built ecologies

Lawrence Berkeley National Laboratory, California

January 2024 – August 2024

Research Affiliate

- Enhanced standardized occupant behavior schema (obXML v.1.4) as a part of the ASHRAE RP-1815 project
- Updated the obFMU Application Guide and IDF files in the tutorial to enhance the compatibility with the latest version of EnergyPlus (v.24.1.0)
- Converted C++-based occupant behavior simulator (obFMU) into Python version for integration with data-driven occupant behavior modeling and BIM
- Explored development of a web platform for real-time energy simulation using EnergyPlus, obFMU, and sensor data
- Hosted by Dr. Tianzhen Hong

Autodesk, New York

May 2023 – August 2023

Intern, Sustainable Active and Passive Mechanical Systems Simulation and Optimization

- Explored integration of generative design process with natural ventilation simulation for evaluating the performance of the ventilation systems at the early design stage using EnergyPlus
- The internship project was published and introduced on the Autodesk Research blog [Link]
- Volunteered to support AEC Mentor Program for underrepresented-minority teenagers who participated in a paid two-week internship at Autodesk's Boston office

Saman Corporation, South Korea

February 2021 – July 2021

Researcher

- Planned application services using real-time data collected from IoT sensors on construction sites in a national research project 'Development of and digital twin platform for smart construction'
- Assisted to prepare kick-off meetings and presentations in a national research project 'XR Flagship Project (Construction part)'
- Wrote bills of quantities in a pilot design project on a box culvert organized by Korea Expressway Corporation

Building Informatics Group, Yonsei University**February 2018 – January 2021***Research Assistant*

- Proposed a technical specification framework for 3D concrete printers through a systematic literature review and a Delphi study
- Developed a multi-dimensional classification algorithm for housing defects text data using BERT-based multitask learning technique (Recurrent Neural Network)
- Explored optimization algorithm for buildability of 3D concrete printing focused on operation planning and concrete properties

ATEAM Ventures, South Korea**May 2015 – August 2015***Strategic Planning staff*

- Planned strategies for marketing & sales and held a 3D-printer exhibition at KINTEX, Ilsan
- Installed a factory to assemble 3D-printers and managed mechanical components
- Managed all customer inquiries, wrote customer manuals, and sent thank-you letters to early customer

XbarX (X_X), South Korea**May 2014 – February 2015***Sales Director & Co-founder*

- Launched a beer recommendation app and beer-distribution web platform
- Got partnerships with 218 beer pub, bar, and restaurant in Seoul, South Korea
- Got a shared 3rd award with a \$4,500 cash prize in 'The 3rd Chung Ju-young Start-up Competition' and funding, \$140,000 from a start-up accelerator

TEACHING
& MENTORING
EXPERIENCE**CodePath, USA****July 2025 – Present***Mentor*

- Advised U.S. college students on application strategies and interview preparation for positions at tech companies at 2025 CodePath's Mentorship & Resume Review Weeks

Mentors in Tech, USA**July 2025 – Present***Mentor*

- Advised U.S. college students on application strategies and interview preparation for software engineering positions at tech companies

Rensselaer Polytechnic Institute, New York**August 2021 – February 2025***Guest Lecturer*

- 'Node-RED 101 for your own Smart IoT-based System' in 'Internetworking of Things (ECSE-4660/ECSE-6660)' course (February 23, 2024)

Project Mentor

- Taught and led an undergraduate group in a research-oriented class 'Research Investigation (ARCH-4958)' (Fall 2022, Fall 2023 & Fall 2024)
- Advised an undergraduate to develop an algorithm evaluating plants' health using a color sensor as a part of the Plant Pixel project sponsored by OBMI (Fall 2023)
- Provided advice to undergraduates in an interdisciplinary project 'Smart Dormitory Devices and Development System (ECSE-4900)' (Fall 2022, Spring 2023 & Fall 2023)
- Participate in Smart City Hackathon 2023 organized by Tech Valley Center of Gravity as a microcontroller-knowledgeable mentor (October 13-15, 2023)

Workshop Instructor

- 'Build an IoT Monitoring System for Your Smart Home' at Black Families Technology Awareness Day (February 2022, February 2023 & February 2025)

Microsoft TEALS, New York
Programming Teaching Assistant

August 2023 – May 2025

- AP Computer Science Principles (Fall 2023, Fall 2024, and Spring 2025) at Franklin D Roosevelt High School, NYC
- AP Computer Science A (Spring 2024) at El Camino High School, San Francisco, CA

Lawrence Berkeley National Laboratory, California
Teaching Assistant & Project Facilitator

June 2024 – July 2024

- Intro to Python & Data Science course in Experiences in Research (EinR) bootcamp 2024 (6-week summer internship)
- Intro to Python & Data Science course in Berkeley Lab Director's Apprenticeship Program (BLDAP) 2024
- Robotics projects in Science Accelerating Girls' Engagement in STEM (SAGE) summer camp 2024

Code Nation, New York
Programming Teaching Assistant

August 2022 – June 2024

- Intro to Web Development (Spring 2024) course at Collegiate Institute for Math & Science, NYC
- Intro to Web Development (AY2022–2023) & Fellowship I (JavaScript/jQuery, Fall 2023) courses at Achievement First East Brooklyn High School, NYC

Yonsei University, South Korea
Teaching Assistant

March 2019 – June 2020

- Engineering-Information Processing (Spring 2020)
- Computer Lab Manager (Spring 2019)

PUBLICATIONS

(Submitted) Chung, J., Huang, S., Liu, B., Lian, J., Dong, J., Katipamula, S. and Lutes, R. (2026) "An Open-Source and Scalable Software Framework for Smart Energy Management in Small and Medium-Sized Commercial Buildings" *Journal of Building Performance Simulation*

(Accepted) Chung, J., Huang, S., Lian, J., Katipamula, S. and Lutes, R. (2026) "Simulation-Based Validation of An Open-Source, Scalable Framework for Building Energy Management in Small and Medium-Sized Commercial Buildings" *Proceedings of SimBuild Conference 2026*

(In Progress) Chung, J., Hong, T., Malik, J. and Shelden, D. (2026) "Enhanced Library of Occupant Behavior Models and gb-obXML Schema Editor for Interoperability between Building Information Modeling and Building Energy Modeling" *Building Simulation*

Ma, N., Lu, Y., Arsano, Y., Brown, N., **Chung, J.**, et al. (2025) "Towards high-performance buildings for thermal resilience and health" *Energy and Buildings*, 116367, <https://doi.org/10.1016/j.enbuild.2025.116367>

Chung, J., Hong, T., Malik, J. and Shelden, D. (2025) "Enhancing Occupant Behavior Representation for Interoperability between Building Information Modeling and Building Energy Modeling" *Building Simulation*, 18(7), 1-24, <https://doi.org/10.1007/s12273-025-1313-z>

Chung, J., Neo, K., Khan, M., Cruz, E., Zheng, Z., Munasinghe, T., and Wei, J. C. (2025) "Exploring Flooded Fraction Prediction through Machine Learning Models Focusing on Medical Infrastructure in the Southeast U.S. Coastal Areas" *Proceedings of the IEEE BigComp 2025*, <https://ntrs.nasa.gov/citations/20250000362>

Chung, J. and Shelden, D. (2024) "A Framework of ifcJSON-based Digital Twin Platform for Monitoring Building Environment using BIM, IoT, and Semantic Web Technologies" *Proceedings of the 20th International Society for Computing in Civil and Building Engineering*, Montreal, Canada, 26 Aug, https://doi.org/10.1007/978-3-031-84208-5_4

- Chung, J.**, Karlicek, B., and Shelden, D. (2024) "Towards Real-Time Occupant Behavior Monitoring System: A Preliminary Study on Integrating BIM, IoT Sensors, and BAS" *Proceedings of ASCE International Conference on Computing in Civil Engineering 2024*, Pittsburg, USA, 29 Jul.
- Chung, J.**, Shahmansouri, N., Stoddart, J., Goldstein, R., and Locke, J. (2024) "Sustainability through Optimal Design of Buildings for Natural Ventilation using Updated Comfort and Occupancy Models" *Proceedings of SimBuild Conference 2024*, 11, pp. 849-860, Denver, USA, 30 May, https://publications.ibpsa.org/conference/paper/?id=simbuild2024_2172
- Wei, J., Gerasimov, I., Munasinghe, T., **Chung, J.**, Cruz, E., Neo, K., Yu, Y. Z., Zheng, Z., and Khan, M. (2024) "Advancing Open Science in Atmospheric Research: Integrating Data Usability and Machine Learning" In *Asia Oceania Geosciences Society (AOGS) 2024 21th Annual Meeting*, <https://ntrs.nasa.gov/citations/20240007793>
- Chung, J.**, Hong, T., Karlicek, R., Santos, H., Shelden, D., and Sparks, D. (2024) "Distributed Semantics to Support Built Environment Digital Twins" In *NSF Workshop on Sustainable Computing for Sustainability 2024*, Alexandria, USA, 17 Apr.
- Chung, J.**, Tsamis, A., and Shelden, D. (2023) "A framework for monitoring and identifying indoor air pollutants based on BIM with IoT sensors" *Proceedings of Computer-Aided Architectural Design. INTERCONNECTIONS: Co-computing beyond boundaries: 20th International Conference, CAAD Futures 2023*, pp. 518-531, Delft, Netherlands, 5 Jul., https://doi.org/10.1007/978-3-031-37189-9_34
- Chung, J.**, Jacoby-Cooper, G., Rook, K., Henrique, S., Shelden, D., Kendall, E., and McGuinness, D. (2023) "Towards an Indoor Environmental Quality Management Ontology" *Proceedings of First International Workshop on Semantic Web on Constrained Things @ ESWC-23 (SWoCoT-23)*, pp. 16-26, Hersonissos, Greece, 28 May, <https://hdl.handle.net/20.500.13015/6674>
- Chung, J.**, Lee, G., and Kim, J. (2021) "Framework for Technical Specifications of 3D Concrete Printers" *Automation in Construction*, 127, 103732, <https://doi.org/10.1016/j.autcon.2021.103732>
- (Submitted) Woo, J., **Chung, J.**, and Koo, H. (2021) "A Study on the Pedestrian Route Choices and Accessibility in the Retail Areas after Acute COVID-19: Focused on the Peak Commercial Activity's Hours in the Mangwon District" *Journal of the Urban Design Institute of Korea*
- Chung, J.**, Lee, G., Kim, J. and Choi, J. (2020) "A Comparative Analysis of the Classification System for Three-Dimensional Concrete Printers" *Korean Journal of Construction Engineering and Management*, 21(2), 3-14, <https://doi.org/10.6106/KJCEM.2020.21.2.003>
- Chung, J.**, Lee, G., and Kim, J. (2020) "A Systematic Review of the Geographic and Chronological Distributions of 3D Concrete Printers from 1997 to 2020", *Proceedings of the 37th International Symposium on Automation and Robotics in Construction (ISARC)*, pp. 84-19, Kitakyushu, Japan, 27-28 Oct., <https://doi.org/10.22260/ISARC2020/0014>
- Lee, G., Cho, J., Song, T., Roh, H., Jung, J., **Chung, J.**, Yong, G., and Jeong, D. (2020) "Construction Field Management Using a Popular Text Messenger" *Proceedings of the 18th International Conference on Computing in Civil and Building Engineering*, pp. 971-979, https://doi.org/10.1007/978-3-030-51295-8_67
- Chung, J.**, Lee, G., and Kim, J. (2018), "Performance Evaluation Criteria for Building 3D Printing Focused on Concrete Properties", *Proceedings of the 2018 Annual Conference of the Korea Institute of Construction Engineering and Management*, pp. 27-29, South Korea, 9 Nov.

PATENTS

Chung, J., Stoddart, J., Locke, J. H., Goldstein, R., and Shahmansouri, N. (2025) “Generation and Simulation of Building Designs for Energy and Comfort Performance”, Application Serial No. US20250238578A1, United States, <https://patents.google.com/patent/US20250238578A1/en>

Chung, J., Stoddart, J., Locke, J. H., Goldstein, R., and Shahmansouri, N. (2025) “Generation and Simulation of Building Designs for Energy and Comfort Performance”, Application Serial No. US20250238566A1, United States, <https://patents.google.com/patent/US20250238566A1/en>

Kim, J., Lee, G., Lee, D., Lee, J. and **Chung, J.** (2020), “3D printer for construction with continuous printing on the go and 3D printing system comprising it”, Application Serial No. KR20200101482A, South Korea, <https://patents.google.com/patent/KR20200101482A/ko>

ARTICLE

Chung, J. (2024) "Want To Be More Comfortable Indoors? Let Your Building Help" *Every Day Matters: RPI Blog*, [Link]

Lee, G., **Chung, J.**, and Kim, J. (2021) "On What Basis Should We Buy a 3D Concrete Printer?" *Building Construction*, 21(4), 31-34, [Link]

HONORS & AWARDS

Faculty Graduate Recognition Award for Excellent Academic Achievement May 2025
Rensselaer Polytechnic Institute, New York

DB Dream Master Scholarship Award for two semesters May 2024
DB Kim-Jun-Ki Cultural Foundation, South Korea

IBPSA-USA SimBuild Conference Student Scholarship Award March 2024
IBPSA-USA SimBuild Conference 2024, United States

Graduate Research Symposium (Podium Presentations), The 3rd Prize June 2022
Rensselaer Polytechnic Institute, New York

Research Assistantship Award for 2021-2025 academic year August 2021
Rensselaer Polytechnic Institute, New York

The 17th Outstanding Thesis Competition, Honorable Mention April 2021
Architectural Institute of Korea, South Korea

2019 Yonsei Startup Lab Contest (\$2,700), Best Startup Award July 2019
Yonsei University, South Korea

2018 KICEM Annual Conference, Excellent Paper Award November 2018
Korea Institute of Construction Engineering and Management, South Korea

Academic Scholarships for four semesters March 2016 – September 2018
Yonsei University, South Korea

The 12th Architectural Engineering Competition, Excellent Award September 2016
The Korean Structural Engineers Association, South Korea

Volunteer Full-Scholarship for two semesters March & September 2016
DB Kim-Jun-Ki Cultural Foundation, South Korea

The 3rd Chung Ju-young Startup Competition (\$4,500), Excellent Startup Award August 2014
The Asan Nanum Foundation, South Korea

The 18th National Boxing Competition for Non-Athlete, The 1st Prize August 2012
Korea Boxing Institution, South Korea

Chief Superintendent's Award for Leadership May 2011
Seoul Metropolitan Office of Education, South Korea

RESEARCH
PROJECTS

Plant Pixel
OBM International, United States

August 2023 – December 2023

**Development of Digital Platform and Digital Twin
based on Management Technology for Smart Construction**
Korea Agency for Infrastructure Technology Advancement (KAIA) [Link]

February 2021 – July 2021

XR Flagship Project (Construction part)
Ministry of Science and ICT & National IT Industry Promotion Agency, Korea [Link]

June 2021 – Jul 2021

**Development of Innovative Design, Material and
Equipment for 3D Printing Small Buildings/Freeform Members**
Korea Agency for Infrastructure Technology Advancement (KAIA) [Link]

March 2018 – February 2021

PRESENTATIONS
& INVITED TALKS

Digital Twin: Multidisciplinary Co-Simulation Ecosystem
GRIDD Lunch & Learn, École de technologie supérieure, University of Québec

December 3, 2024

**A framework of BIM & IoT-based Building Infrastructure
for Monitoring Indoor Air Quality and Occupant Behavior**
Research Showcase — Reunion & Homecoming 2024, RPI [Poster]

September 27, 2023

**IFC Simplification and Granular Component Demo
& IFC-LD / SHACL Implementation Developments**
General Assembly of Implementers, buildingSMART International [Oral]

August 29, 2024

How Programming is Used in Architecture & Building Technology?
Computer Science Career Panel, El Camino High School

April 11, 2024

**A Daily Flooding Risk Assessment Map for Medical Infrastructures
Using Machine Learning Algorithms**
AGU23 Fall Meeting [Poster]

January 22, 2024

**A framework of BIM & IoT-based Building Infrastructure
for Monitoring Indoor Air Quality and Occupant Behavior**
Research Showcase — Reunion & Homecoming 2023, RPI [Poster]

October 13, 2023

**Sustainability through Optimal Design of Buildings
for NV and Post-Occupancy Using Updated Comfort Models**
Research Connections — Summer Intern Series, Autodesk [Oral]

August 17, 2023

**Daily Flooding Risk Assessment Map for Medical Infrastructures
using Machine Learning Algorithm**
AmeriGEO Week 2023 'Data Driven Solutions for a Sustainable Planet' [Poster]

August 8, 2023

Towards an Indoor Environmental Quality Management Ontology
2023 8th Annual Graduate Research Symposium, RPI [Oral]

May 12, 2023

**A framework for monitoring and identifying indoor air pollutants
based on BIM with IoT sensors**
2023 8th Annual Graduate Research Symposium, RPI [Poster]

May 12, 2023

Air Quality Monitoring System based on BIM & IoT
EBESS Industry Workshop 'Well-being & Sustainability in the Built Environment', RPI [Poster]

November 2, 2022

Air Quality Monitoring System based on BIM & IoT
2022 7th Annual Graduate Research Symposium, RPI [Poster, Oral – 3rd Place]

May 13, 2022

A Systematic Review of 3D Concrete Printers from 1997 to 2020
37th International Symposium on Automation and Robotics in Construction (ISARC) [Oral]

October 28, 2022

SKILLS

Programming Languages: Python, C#, JAVA, C++, JavaScript (React.js, Node.js, Three.js, ifc.js, Express.js, Node-RED), TypeScript, P5 Processing, SPARQL, SQL
CAD/CAE: Rhino (+Grasshopper, Ladybug, Honeybee, Butterfly), Revit (+Dynamo, Ladybug, Honeybee, Butterfly), Sketchup, CATIA, AutoCAD, Midas Civil
Design tools: Adobe Photoshop, Illustrator, InDesign, Premiere Pro, Lightroom, Experience Design
Machining: Arduino, Raspberry Pi, 3D printer, Laser Cutter
Others: OpenStudio (+EnergyPlus), Contam, Functional Mockup Unit (+obFMU, EnergyPlusFMU, ContamFMU), Tensorflow (+Keras), Pytorch, SPSS, SHACL, Protégé

CERTIFICATIONS

LEED AP BD+C (Credential ID: 11476216-AP-BD+C) <i>U.S. Green Building Council</i>	September 20, 2024
Facility Management Professional (Credential ID: 1193106) <i>International Facility Management Association (IFMA)</i>	June 27, 2024
WELL AP (Credential ID: WELL-AP-0000134489) <i>International WELL Building Institute (IWBI)</i>	February 6, 2023
LEED Green Associate (Credential ID: 11476216-GREEN-ASSOCIATE) <i>U.S. Green Building Council</i>	August 13, 2022
2020 AI Technical Development Workshop (Image/Video Recognition) <i>Korea Electric Power Corporation (KEPCO)</i>	August 7, 2020
The 9th Image Recognition Application Workshop <i>Fast Campus (패스트캠퍼스)</i>	October 19, 2019
The 3rd Object Detection Application Workshop <i>Fast Campus (패스트캠퍼스)</i>	September 7, 2019
DA BIM Specialist Intermediate Workshop <i>DAGROUP Urban Design & Architecture Co., Ltd</i>	July 14, 2019
2018 BIGDATA X CAMPUS (Python-based Big Data Analysis) <i>Korea Data Agency</i>	June 14, 2018

**ACADEMIC
SERVICE**

Journal Reviewer

- Automation in Construction
- Building Simulation: An International Journal
- Journal of Building Engineering
- Buildings
- Energies
- Sustainability
- Processes
- Archives of Advanced Engineering Science
- Journal of Data Science and Intelligent Systems
- Journal of Civil Engineering and Architecture
- Journal of Civil, Construction and Environmental Engineering

Academic Conference Reviewer

- IBPSA-USA SimBuild Conference 2024 / 2026
- IBPSA-Australasia Building Simulation Conference 2025
- International Conference on Modern Management based on Big Data 2025
- International Conference on Artificial Intelligence, Computer, Data Sciences and Applications 2025
- International Conference on Electrical, Computer and Energy Technologies 2025
- International Conference on Electrical, Computer, Communications and Mechatronics Engineering 2024 / 2025
- International Conference on Electrical and Computer Engineering Researches 2024

Invited Juror

- Architectural Drawing & Representation at GSAPP, Columbia University (Fall 2023)
- 2025 Shape the Future of Building Simulation Competition by IBPSA-USA
- 2025 GEM Fellow Tech Talk Competition at Oak Ridge National Laboratory
- Green Scholars Program Science Fair 2024
- Contra Costa County Science & Engineering Fair 2024
- Alameda County Science & Engineering Fair 2024

Application Reviewer

- Science en Accion 2024 at Berkeley Lab
- Science Accelerating Girls' Engagement in STEM (SAGE) 2024 at Berkeley Lab
- Quantum, Computing, Mathematics & Physics (QCaMP) 2024 at Berkeley Lab
- Berkeley Lab Director's Apprenticeship Program (BLDAP) 2024

Session Chair

- IBPSA-USA SimBuild 2024

Committee Member

- Budget & Sponsorship Chair, Executive Committee Board for FY2026, Oak Ridge Postdoctoral Association
- Member, Research Committee, International Building Performance Simulation Association - USA (IBPSA-USA)
- Provisional Corresponding Member, Technical Committee (T-TAC-TC07.05), American Society of Heating, Refrigerating and Air-conditioning Engineers (ASHRAE)

MEDIA APPEARANCES

Meeting Future Talents with Diverse Dreams: **June 11, 2024**

2024 DB Dream Master Scholarship Award Ceremony
DB Story Blog, DB Kim-Jun-Ki Cultural Foundation, Korea [Link]

Recently Published by Autodesk Researchers **January 18, 2024**
Autodesk Research Blog [Link]

CASE Doctoral Student To Conduct Fellowship at Berkeley Lab **January 10, 2024**
Every Day Matters, Rensselaer Polytechnic Institute [Link]

Nurturing Blue Dreams at the School of Architecture **September 5, 2018**
EBS News, South Korea [Link]

VOLUNTEER EXPERIENCE

Lawrence Berkeley National Laboratory, California **January 2024 – August 2024**
Application Review Board / Mentor / STEM Activity Supporter

Y-Zone Project in STEM Alliance, New York **October 2021 – September 2022**
Bilingual (Korean/English) Volunteer Translator

Major Mentoring Program in Yonsei University, Korea <i>Architecture-Major Mentor</i>	March 2019 – May 2021
Architecture School in Junglim Foundation, Korea <i>Architecture Tutor</i>	August 2018 – October 2018
Global Volunteering Program in Dongbu Foundation, Vietnam <i>Student Volunteer</i>	January 2017
Student Volunteer Group in Dongbu Foundation, Korea <i>Volunteer & Scholarship Holder</i>	February 2016 – February 2017
Global Engineering Program in Yonsei University, Indonesia <i>Student Volunteer & Assistant Manager</i>	August 2016
Seoul Junghwa Secondary School , Korea <i>Speech Tutor</i>	September 2012 – December 2012
Seoul Soongduck Primary School , Korea <i>Speech Tutor</i>	March 2012 – July 2013
Kyoto International School , Japan <i>Korean Tutor</i>	August 2012

REFERENCES

Available upon request.